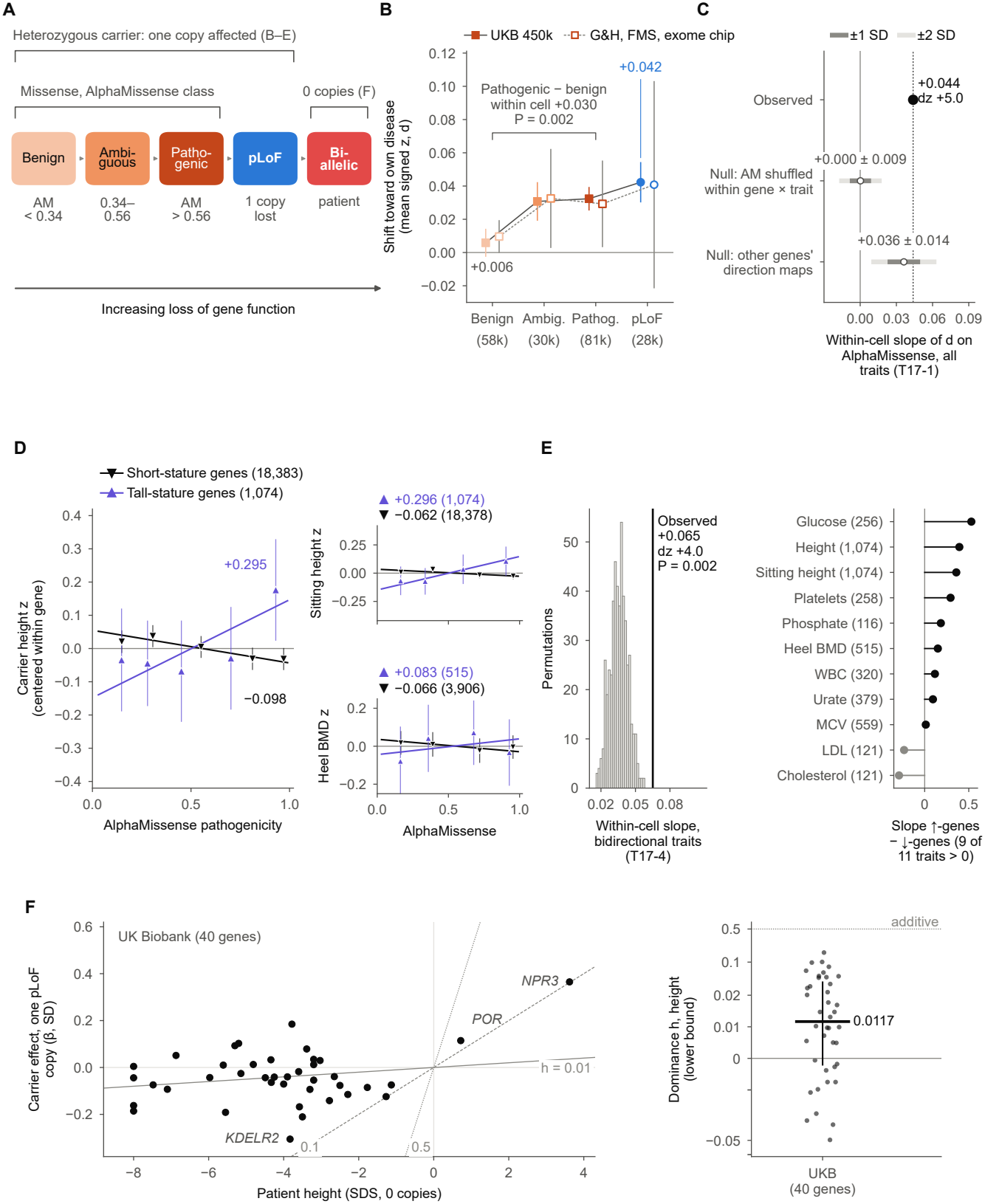

A
Heterozygous carrier: one copy affected (B–E)
Missense, AlphaMissense class
0 copies (F)
Benign
Ambi-guous
Patho-genic
pLoF
Bi-allelic
AM < 0.34
0.34–0.56
AM > 0.56
1 copy lost
patient
Increasing loss of gene function
B
UKB 450k
G&H, FMS, exome chip
Shift toward own disease (mean signed z, d)
Pathogenic − benign within cell +0.030 P = 0.002
+0.042
+0.006
Benign (58k)
Ambig. (30k)
Pathog. (81k)
pLoF (28k)
C
±1 SD
±2 SD
Observed
+0.044 dz +5.0
+0.000 ± 0.009
Null: AM shuffled within gene × trait
+0.036 ± 0.014
Null: other genes' direction maps
Within-cell slope of d on AlphaMissense, all traits (T17-1)
D
Short-stature genes (18,383)
Tall-stature genes (1,074)
Carrier height z (centered within gene)
+0.295
−0.098
AlphaMissense pathogenicity
+0.296 (1,074)
−0.062 (18,378)
Sitting height z
+0.083 (515)
−0.066 (3,906)
Heel BMD z
AlphaMissense
E
Permutations
Observed +0.065 dz +4.0 P = 0.002
Within-cell slope, bidirectional traits (T17-4)
Glucose (256)
Height (1,074)
Sitting height (1,074)
Platelets (258)
Phosphate (116)
Heel BMD (515)
WBC (320)
Urate (379)
MCV (559)
LDL (121)
Cholesterol (121)
Slope ↑-genes − ↓-genes (9 of 11 traits > 0)
F
UK Biobank (40 genes)
Carrier effect, one pLoF copy (β, SD)
NPR3
POR
KDELR2
h = 0.01
0.1
0.5
Patient height (SDS, 0 copies)
Dominance h, height (lower bound)
additive
0.0117
UKB (40 genes)